

Bioinorganic chemistry

Introduction

Role of metal ions in biological systems

O₂ transport in biological systems

Haemoglobin and myoglobin

Haemocyanin

Haemerythrin

Metalloenzymes

Carbonic anhydrase

Carboxypeptidase

ELEMENTAL MASS ABUNDANCE IN A 70 kg HUMAN

ELEMENTAL MASS ABUNDANCE IN A 70 kg HUMAN

Bulk elements

Oxygen	44	kg
Carbon	12.6	kg
Hydrogen	6.6	kg
Nitrogen	1.8	kg
Sulfur	0.1	kg

Macrominerals

Calcium	1700	g
Phosphorus	680	g
Potassium	250	g
Chlorine	115	g
Sodium	70	g
Magnesium	42	g

Trace elements

Iron	5000	mg
Silicon	3000	mg
Zinc	1750	mg
Rubidium	360	mg
Copper	280	mg
Strontium	280	mg
Bromine	140	mg
Tin	140	mg
Manganese	70	mg
Iodine	70	mg
Aluminum	35	mg
Lead	35	mg
Barium	21	mg
Molybdenum	14	mg
Boron	14	mg
Arsenic	~ 3	mg
Cobalt	~ 3	mg
Chromium	~ 3	mg
Nickel	~ 3	mg
Selenium	~ 2	mg
Lithium	~ 2	mg
Vanadium	~ 2	mg

Chemical elements essential to life forms can be divided into the following

(i) Bulk elements: C, H, N, O, P, S

(ii) Macrominerals and ions: Na, K, Mg, Ca, Cl, PO_4^{3-} , SO_4^{2-}

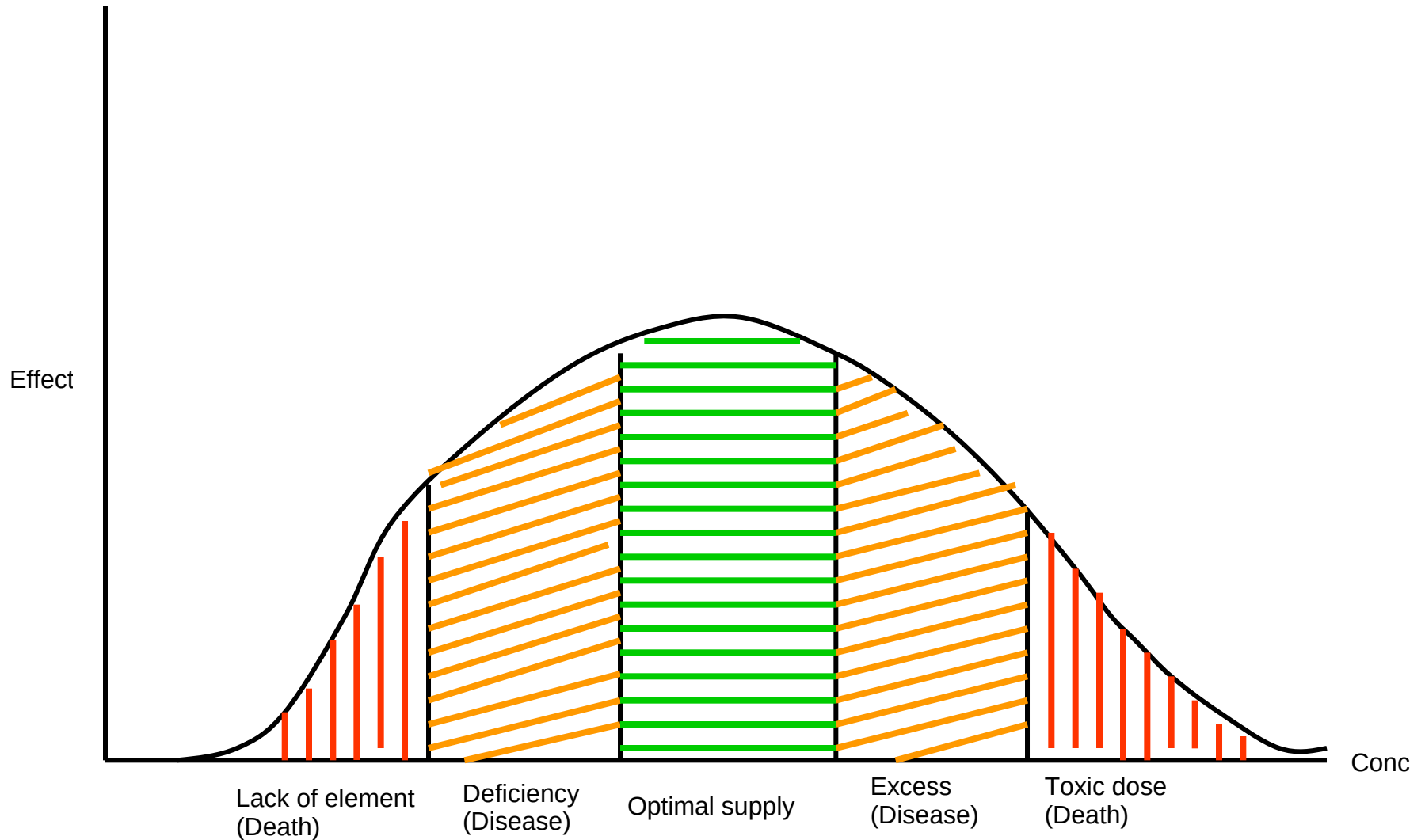
(iii) Trace elements: Fe, Zn, Cu

(iv) Ultratrace elements comprises of

(a) non-metals: F, I, Se, Si, As, B

(b) metals: Mn, Mo, Co, Cr, V, Ni, Cd, Sn, Pb, Li

Concentration and physiological effect



Role of metal ions in biological systems

Na⁺ and K⁺ Regulation of the osmotic pressure, membrane potentials, enzyme activity, signalling

Mg²⁺ Chlorophyll; anaerobic energy metabolism

V^{IV/V}, Mo^{IV/V}, W^{IV/V}, Mn^{II/III/IV}, Fe^{II/III}, Ni^{II/III}, Cu^{I/II} active centres in electron-transport (redox) enzymes, oxygenases, dismutases.

Fe and Cu: Transport of oxygen.

Fe^{III} Iron-storage proteins (ferritins).

Co Synthases and isomerases (cobalamines, e.g. vitamin-B12); methylation of inorganics.

Zn²⁺ In the active centre of hydrolases, carboanhydrase, alcohol dehydrogenase, synthases; genetic transcription (zinc fingers), stabilisation of tertiary and quartary structures of proteins; repair enzymes.

P^V (phosphate): Constituent in hydroxi- and fluorapatite ,energy metabolism (ATP), activation of organic substrate; phospholipids in cell membranes; phosphate esters (DNA, RNA,...).

Li⁺: Treatment of bipolar disorder (maniac depression) and hypertension.

Gd³⁺: Contrast agent in magnetic resonance tomography of soft tissues.

BaSO₄: Contrast agent for X-ray tomography. Sun protection.

Pt^{II}: Chemotherapy (e.g. with cisplatin *cis-[Pt(NH₃)₂Cl₂]*) of cancer (*ovaria, testes*)

Au^I: Therapy of rheumatic arthritis

Sb^{III}: Treatment of inflammatory skin pimples like acne

Bi^{III}: Treatment of gastritis

PROTEINS – CLASSES AND FUNCTIONS

dynamic

catalysis

transport

protection

muscle contraction

metabolic control

gene transcription

storage

enzymes

hemoglobin

antibodies

actin and myosin

hormones

histones

ferritin

structural

matrices for bone
and connective tissue

collagen and elastin

PROTEINS

proteins are polymers of 20 different α -amino acids, known as the common amino acids.

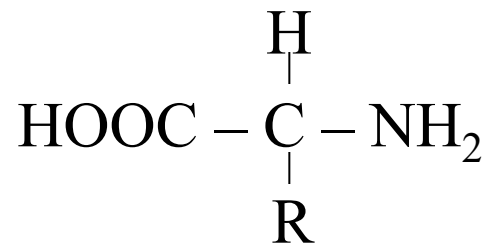
properties of 20 amino acids

α -amino group – except proline, which has an imino group

α -carboxyl group

unique R side chain and a hydrogen bound at the central α carbon

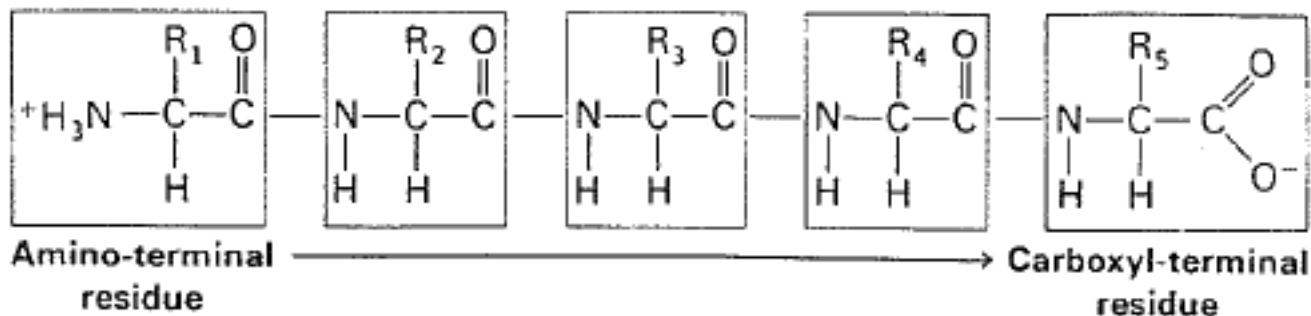
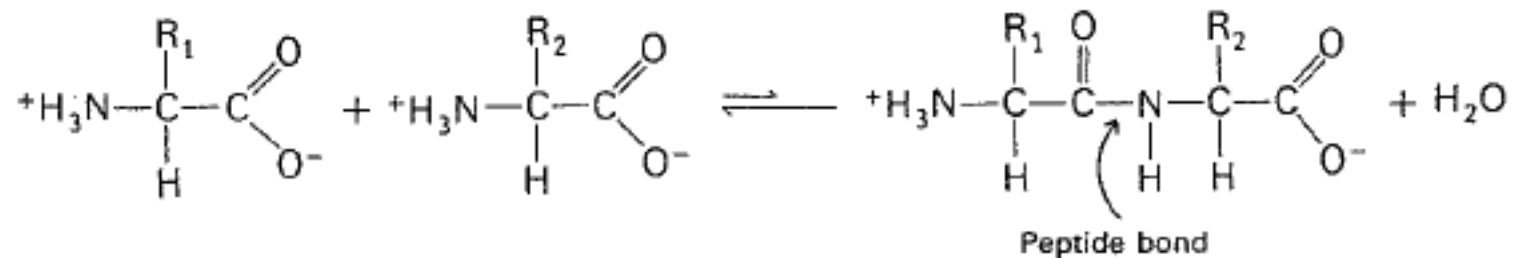
possess at least one asymmetric carbon (L form) except glycine



PROTEINS

at neutral pH, the amino and carboxyl groups are ionized, and the amino acids thus exist as zwitterions

proteins are produced by enzymatic polymerization of the 20 common amino acids, connected by peptide bonds formed by dehydration



AMINO ACIDS – ALIPHATIC

Glycine
(Gly, G) — H

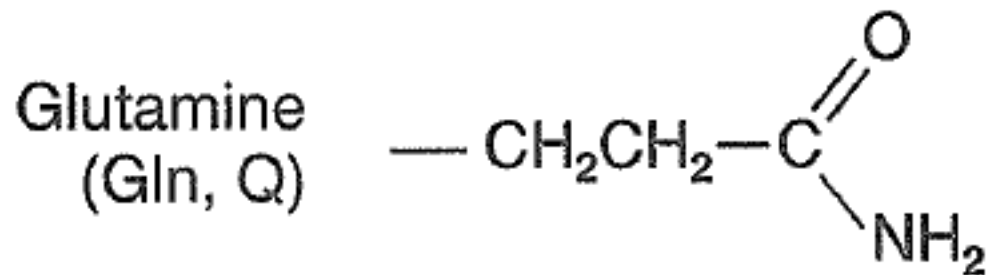
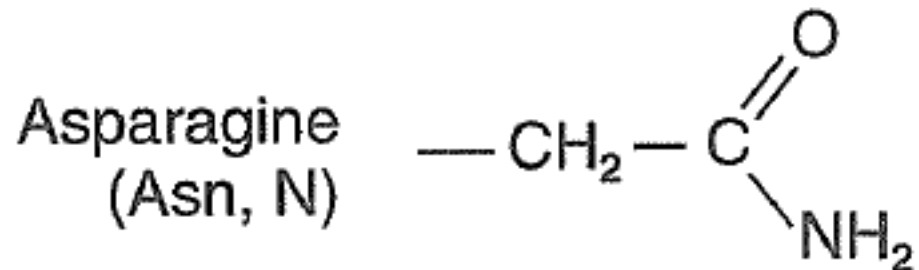
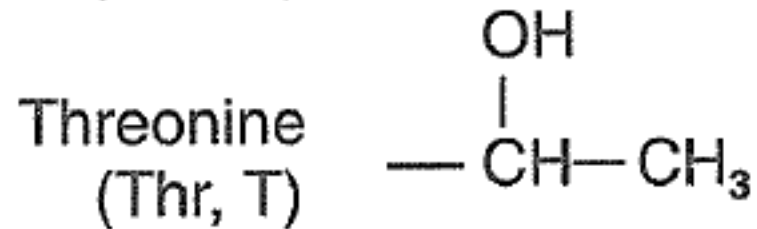
Alanine
(Ala, A) — CH₃

Valine
(Val, V) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{—CH—CH}_3 \end{array}$

Leucine
(Leu, L) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{—CH}_2\text{—CH—CH}_3 \end{array}$

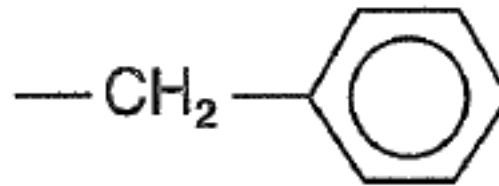
Isoleucine
(Ile, I) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{—CH—CH}_2\text{—CH}_3 \end{array}$

AMINO ACIDS – POLAR

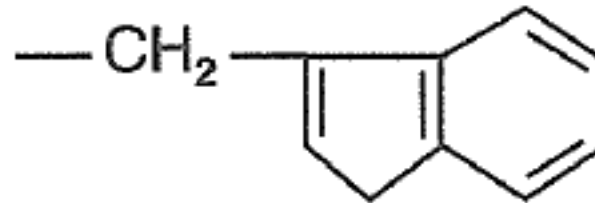


AMINO ACIDS – AROMATIC

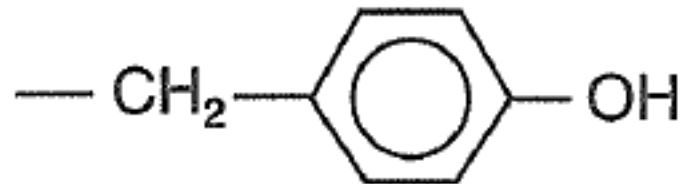
Phenylalanine
(Phe, F)



Tryptophan
(Trp, W)



Tyrosine
(Tyr, Y)



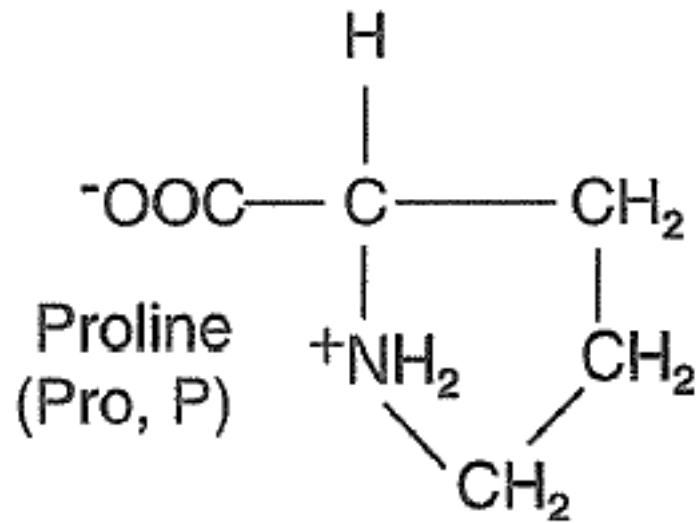
AMINO ACIDS – SULFUR OR SELENIUM

Sulfur-containing

Methionine — $\text{CH}_2\text{CH}_2\text{—S—CH}_3$
(Met, M)

Cysteine — $\text{CH}_2\text{—S—H}$
(Cys, C)

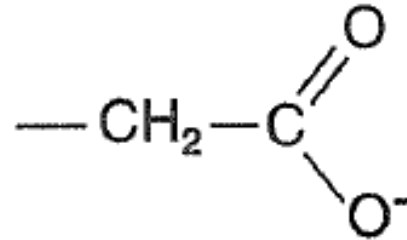
AMINO ACIDS – SECONDARY AMINE



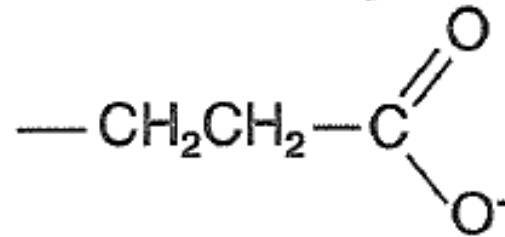
AMINO ACIDS – CHARGED

Acidic

Aspartate
(Asp, D)



Glutamate
(Glu, E)

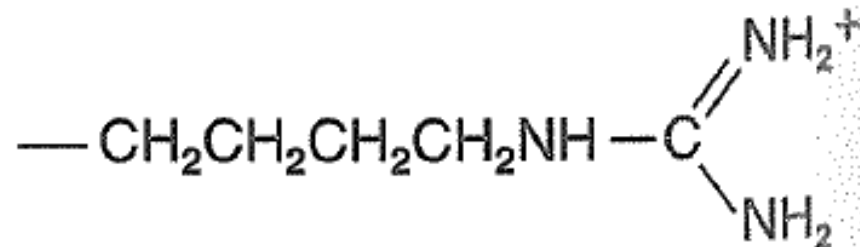


Basic

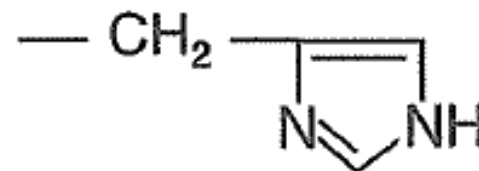
Lysine
(Lys, K)



Arginine
(Arg, R)



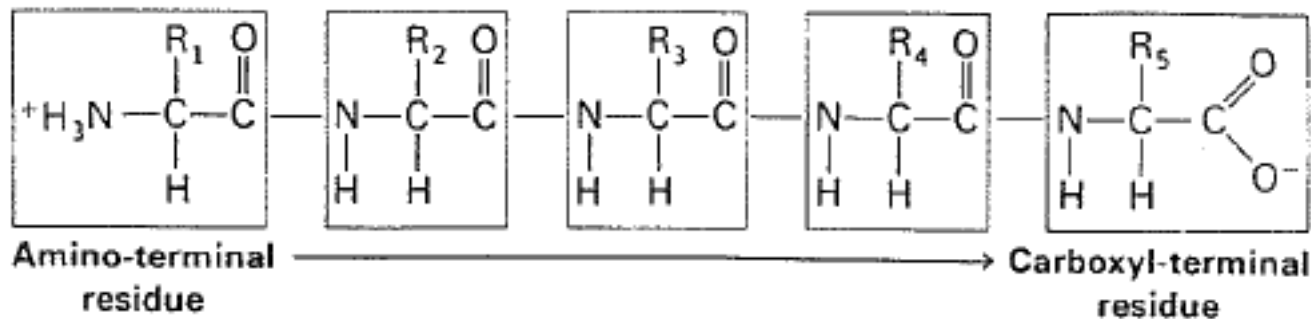
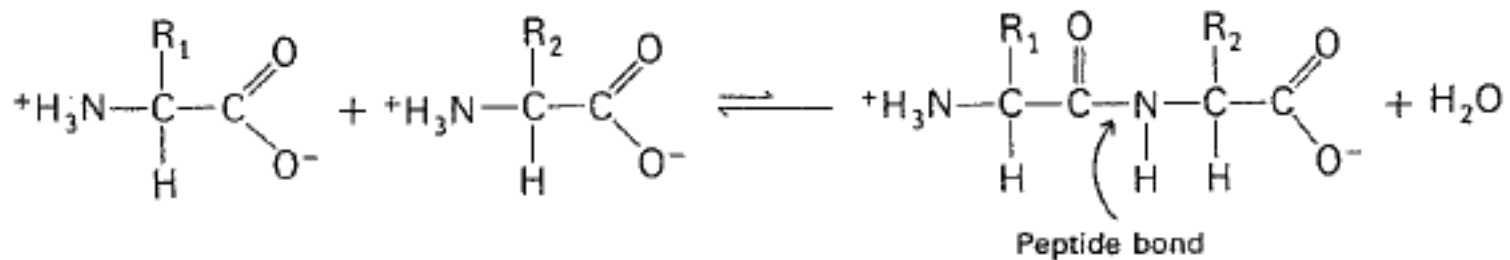
Histidine
(His, H)



PROTEINS

proteins are produced by enzymatic polymerization of the 20 common amino acids, connected by peptide bonds formed by dehydration

the specific sequence of amino acids in the polypeptide chain is called the *primary structure* of the protein.



METAL LIGATION

metal ions are bound in mononuclear or polynuclear coordination units in which amino acid side chains function as multidentate chelating ligands (protein)

common bridging ligands

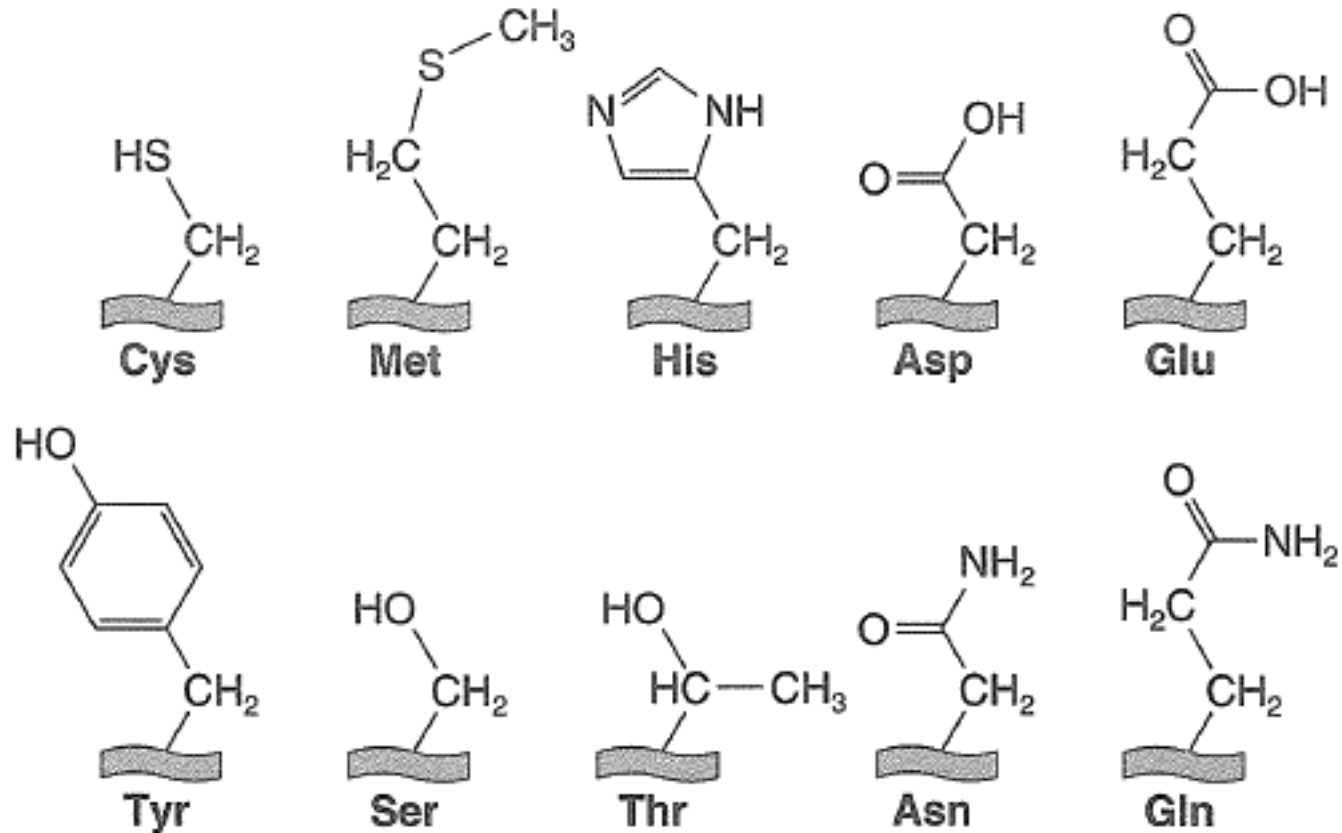
O^{2-} , OH^- , $-\text{CH}_2\text{S}^-$, S^{2-} , $-\text{CH}_2\text{CO}_2^-$, imidazole

exogenous terminal ligands are also often bound to metals

H_2O , OH^- , O^{2-} , HS^- , S^{2-}

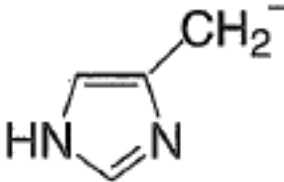
METAL LIGATION

Protein residues as ligands for metal ions



Oxygen atoms of peptide carbonyls, nitrogen atoms of deprotonated backbone amides, and lysine side chains are also available for metal coordination.

METAL LIGATION

R Group (ligand)	Amino Acid	Metals Bound ^a
$-\text{CH}_2\text{SH}^b$ $-\text{CH}_2\text{SeH}^b$	Cys SeCys	$\left\{ \begin{array}{l} \text{Fe}^{2+,3+}, \text{Ni}^{2+,3+} \\ \text{Cu}^{1+,2+}, \text{Zn}^{2+} \\ \text{Cd}^{2+}, \text{Mo}^{4+,6+} \end{array} \right.$
$-\text{CH}_2-\text{C}_6\text{H}_4-\text{OH}^b$	Tyr	$\text{Mn}^{3+}, \text{Fe}^{2+,3+}$
	His	$\text{Fe}^{2+,3+}, \text{Mn}^{2+,3+}, \text{Ni}^{2+}$ $\text{Cu}^{1+,2+}, \text{Zn}^{2+}$
$-\text{CH}_2\text{SCH}_3$	Met	$\text{Fe}^{2+,3+}, \text{Cu}^{1+,2+}$
$-(\text{CH}_2)_{1,2}\text{COOH}^b$	Asp, Glu	$\text{Mg}^{2+}, \text{Ca}^{2+}, \text{Mn}^{2+,3+}$ $\text{Fe}^{2+,3+}, \text{Ni}^{2+}, \text{Zn}^{2+}$
$\begin{array}{c} \rightarrow \text{O} \\ \parallel \\ -\text{HN}-\text{C}-\text{CHR}-\text{NH}- \end{array}$	any	$\text{Cu}^{2+}, \text{Zn}^{2+}, \text{Mg}^{2+}, \text{Ca}^{2+}$

The prosthetic group

It is an additional, non-amino acid component of a protein which is essential for the biological activity of the protein.

We shall be concerned with prosthetic groups containing metal centres,

e.g. **haem** is the prosthetic group in haemoglobin and myoglobin.

PROSTHETIC GROUPS

biosynthesized groups that may participate directly in catalytic processes or carry other small molecules; binding to proteins is strong

functions

- bind metal cations tightly

- may accept a substrate

- may participate in electron transfer

MACROCYCLIC LIGANDS

tetrapyrroles

most common, best known bioinorganic compounds

study of structure/function and organic synthesis of these complexes
led to several Nobel prizes

1915 – **Willstätter** (extraction of pigments, relationship between chlorophyll and heme)

1930 – **Fischer** (formula of heme and chlorophyll, first synthesis of tetrapyrroles)

1962 – **Kendrew & Perutz** (X-ray structure of hemoglobin and myoglobin)

1964 – **Crowfoot Hodgkin** (X-ray structure of vitamin B₁₂)

1965 – **Woodward** (total synthesis of vitamin B₁₂ and chlorophyll)

1988 – **Deisenhofer, Huber, & Michel** (X-ray structure of photosynthetic reaction centers containing heme and chlorophyll in bacteria)

Metalloproteins : The proteins that contain metals

Apoprotein: The form of the protein with the metal removed is called the apoportion.

The prefix apo- before a particular protein (e.g. ferritin and apoferritin) signifies the metal-free species.

The difference between a protein and the corresponding apoprotein is analogous to that between a metal complex and the corresponding free ligand

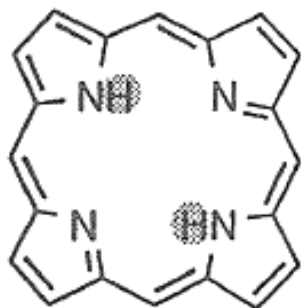
TETRAPYRROLES

partially unsaturated, tetradentate, macrocyclic ligands

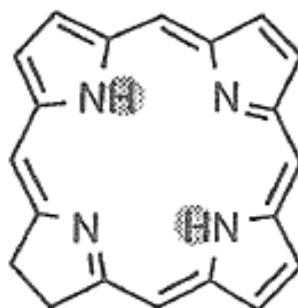
stable, rigid, planar or nearly planar ring system

deprotonated forms bind metal ions tightly and size selectively

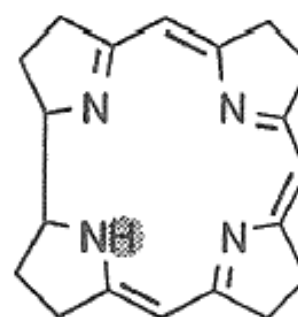
extensive conjugation leads to very intense colors (*pigments of life*) and potentially to redox activity



porphyrin
or 'porphin' (in
unsubstituted form)



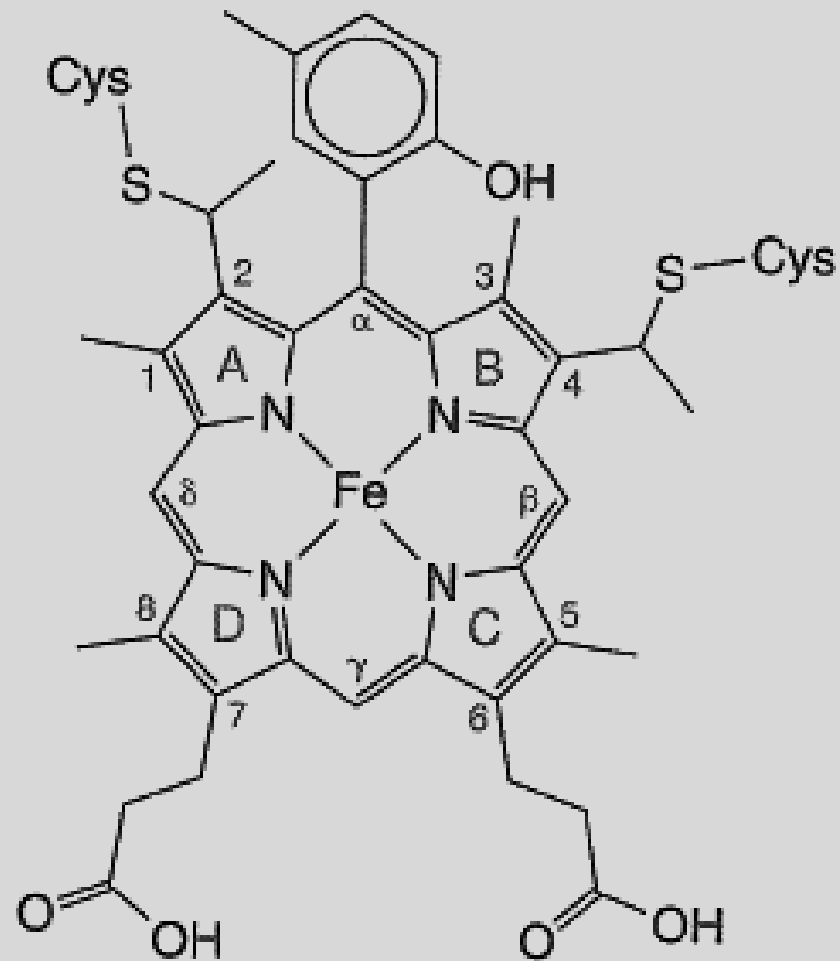
chlorin
(2,3-dihydroporphyrin)



corrin

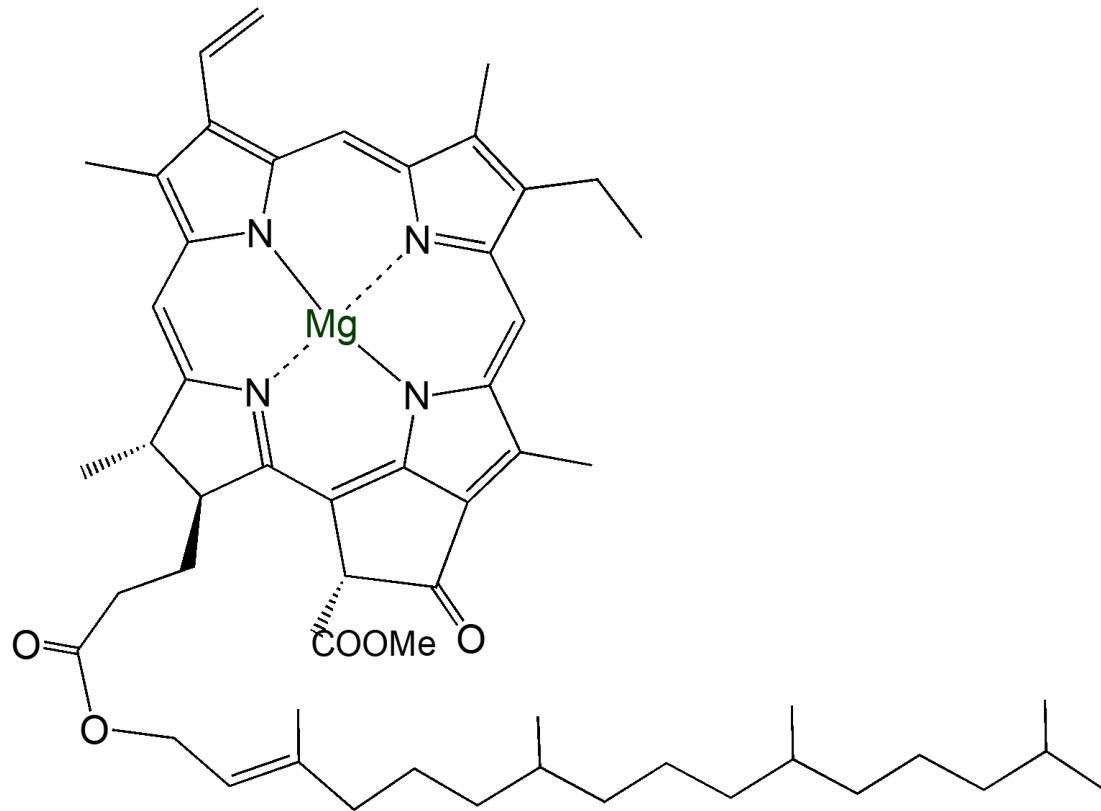
⦿: acidic protons which may be substituted by metal ions

PORPHYRINS



Heme P460

CHLORINS – CHLOROPHYLL *a*

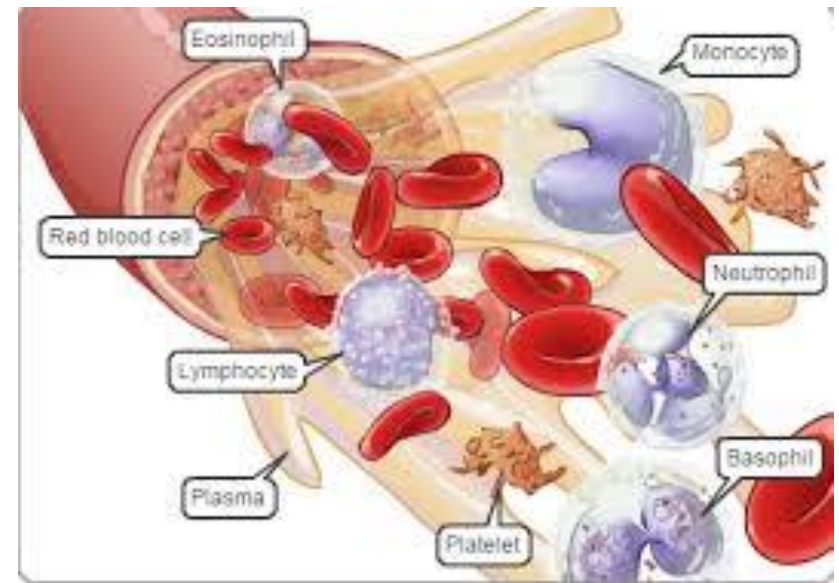
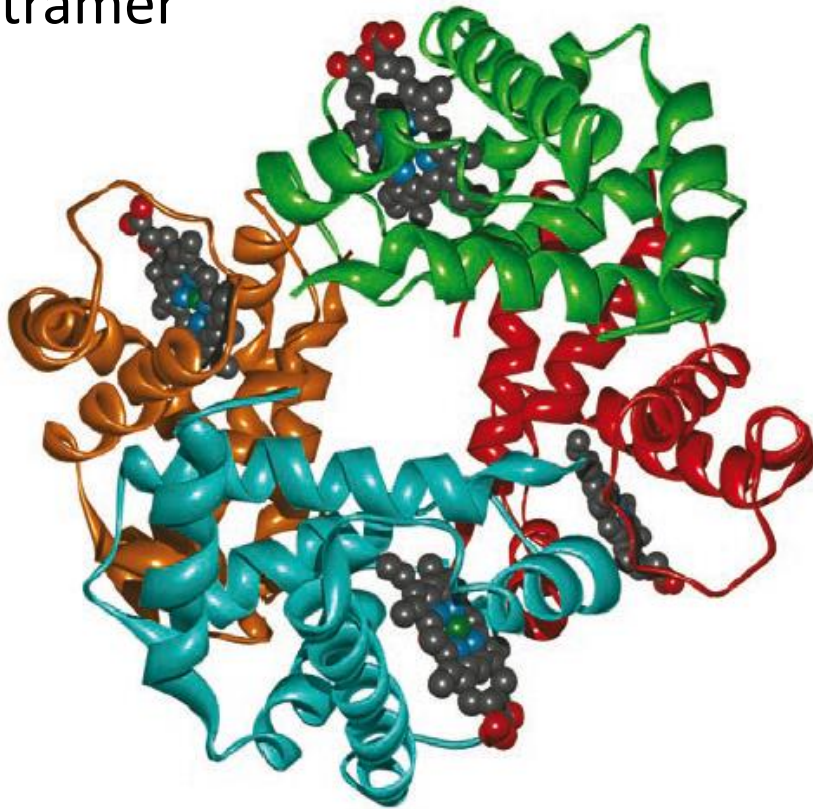


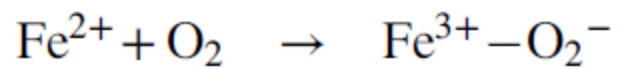
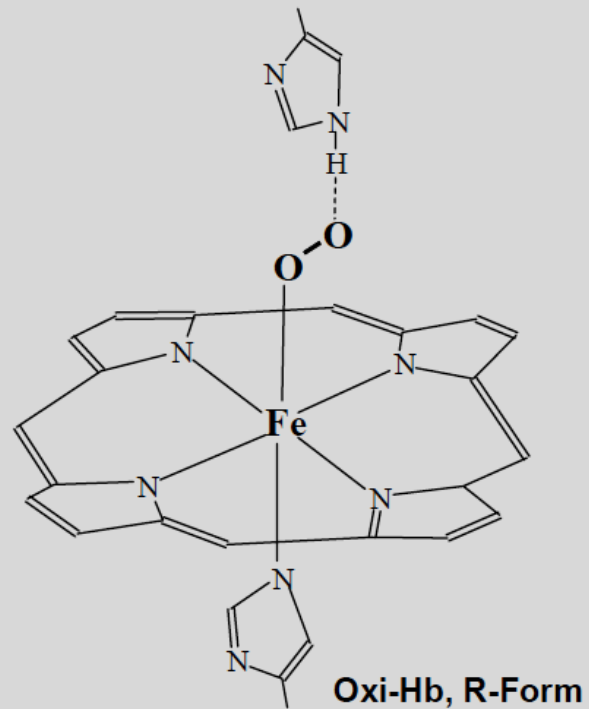
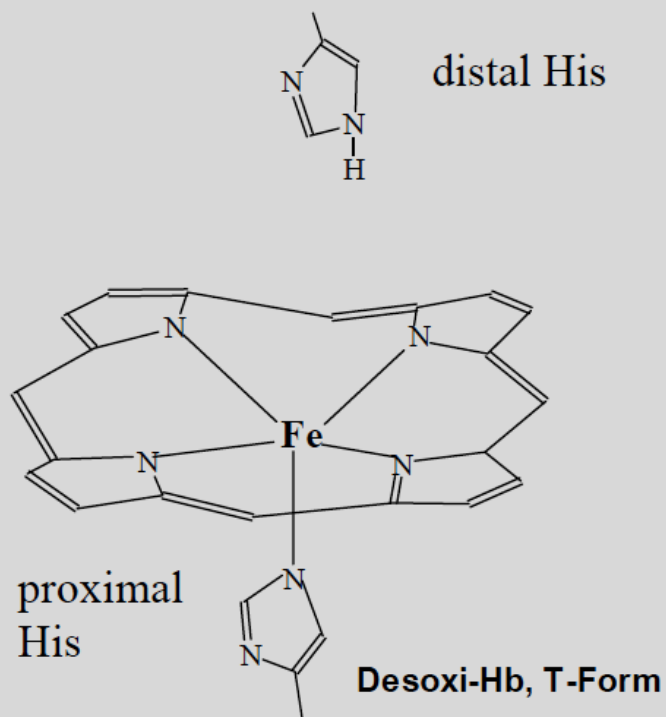
chlorophyll *a*

Haemoglobin and Myoglobin

Myoglobin has a molecular weight of 17000 and is a monomer with a protein chain consisting of 153 amino acid residues.

Haemoglobin has a molecular weight of 64500 and is a tetramer

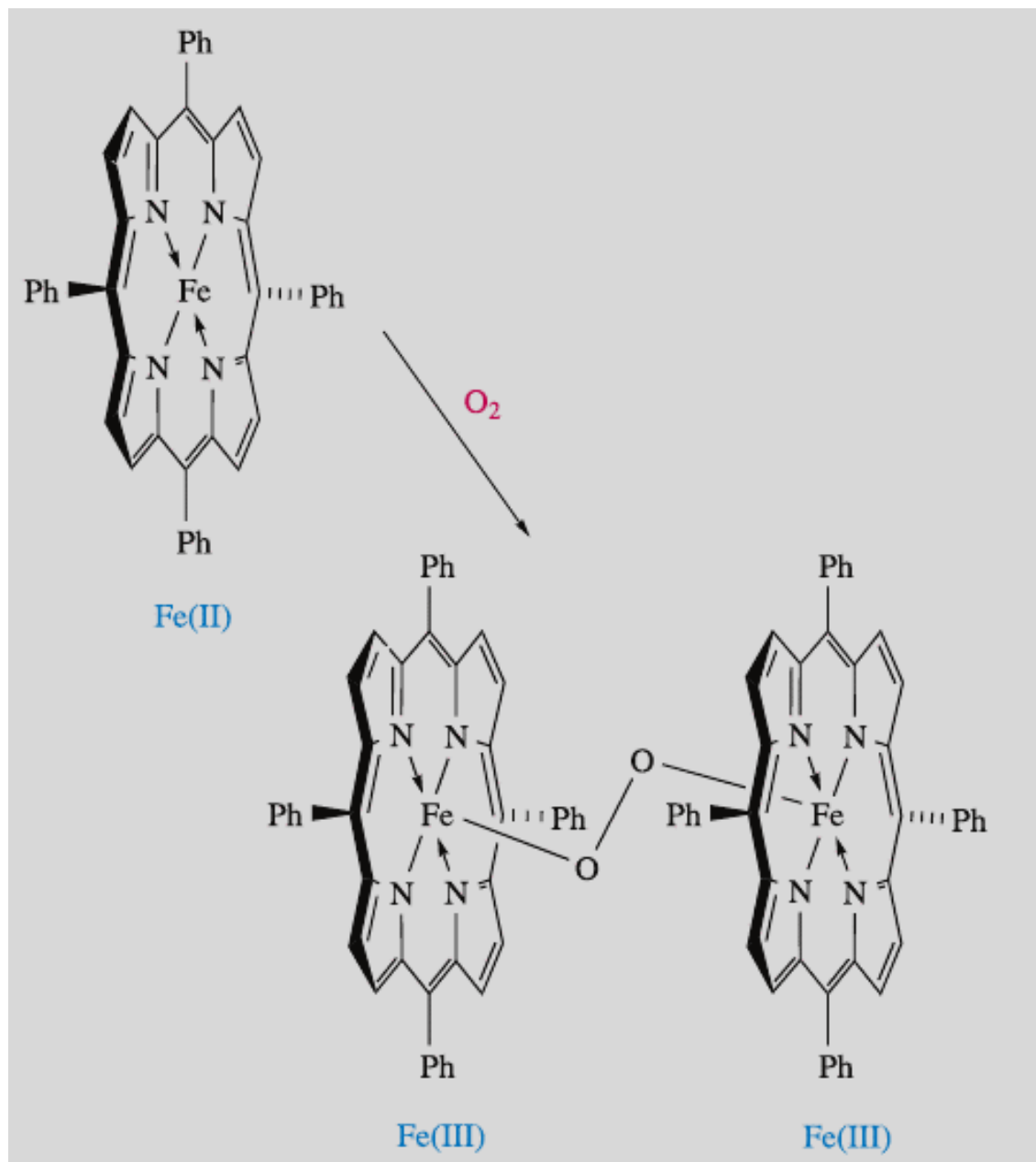




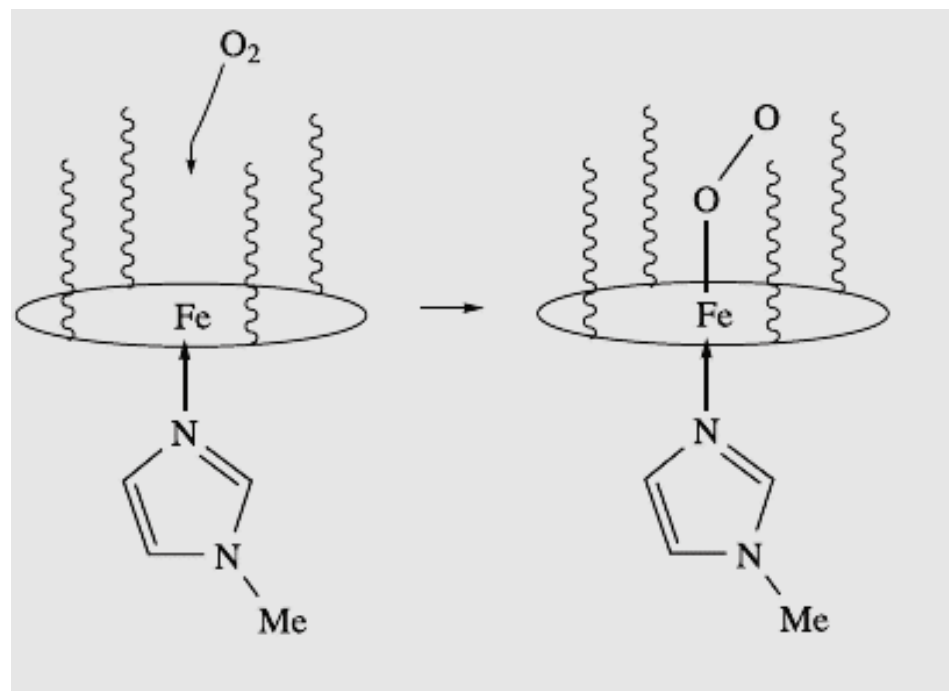
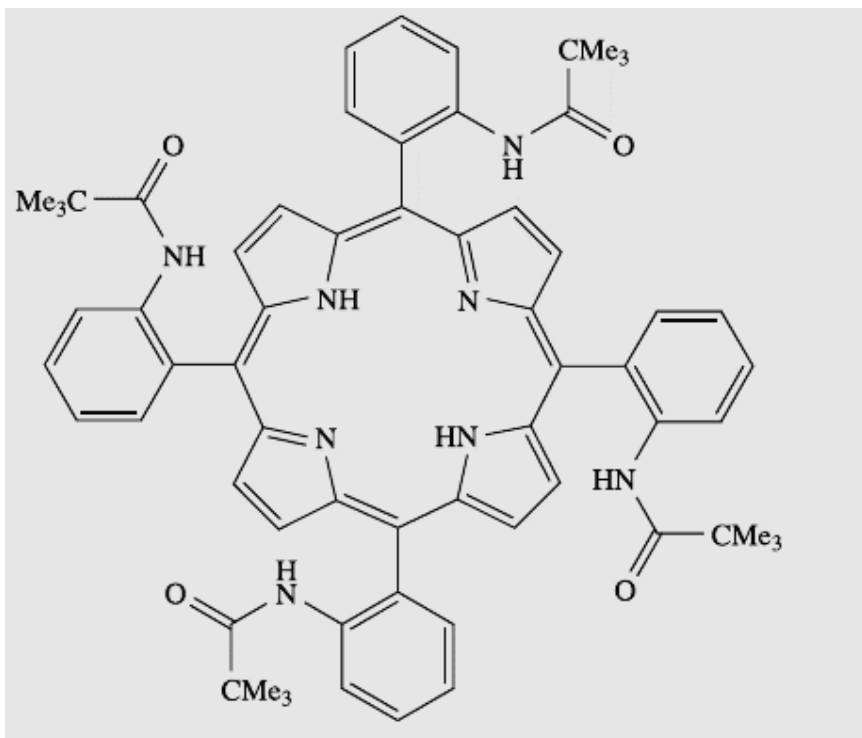
Deoxy-Mb

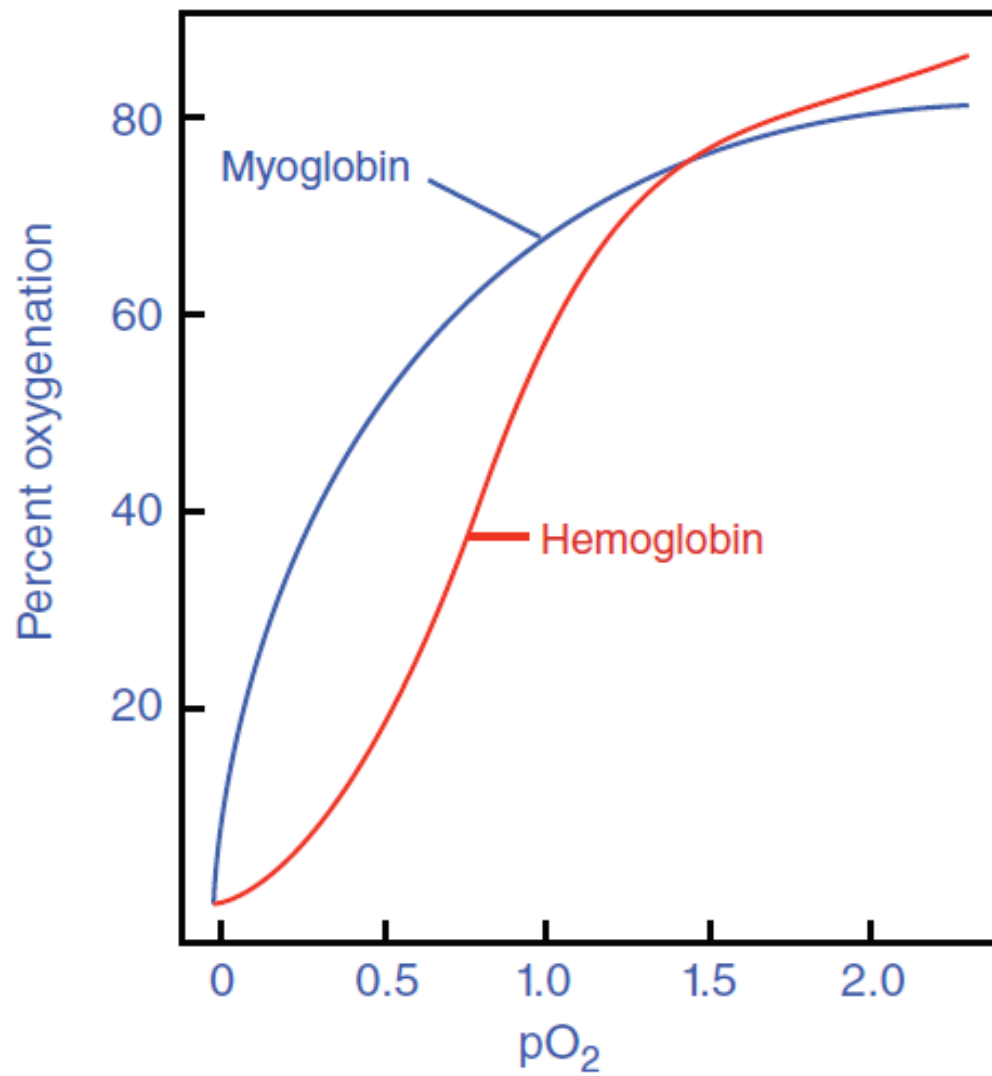
Oxy-Mb,

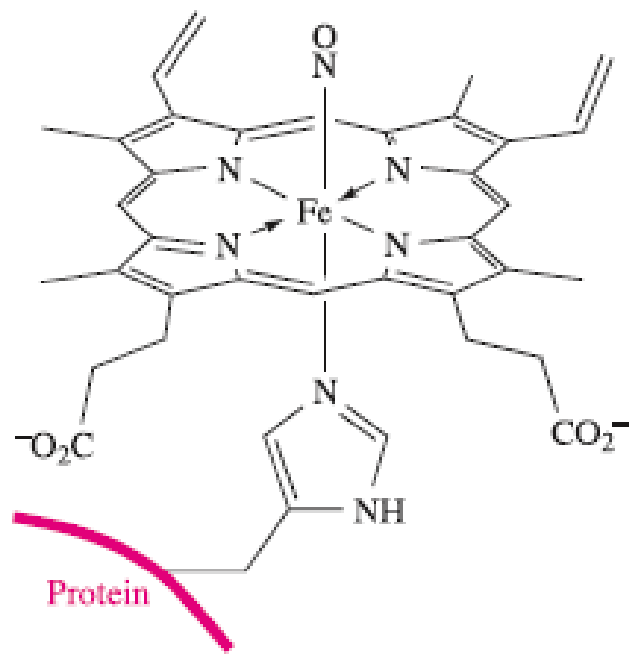
	O_2	O_2^-	O_2^{2-}
ν_{O-O} (cm^{-1})	1560	1150–1100	850–740
Bond length ($\text{\AA}$)	1.21	1.28	1.49
Bond order	2	1.5	1



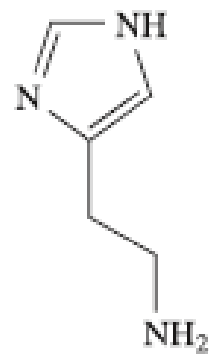
'picket-fence' porphyrin



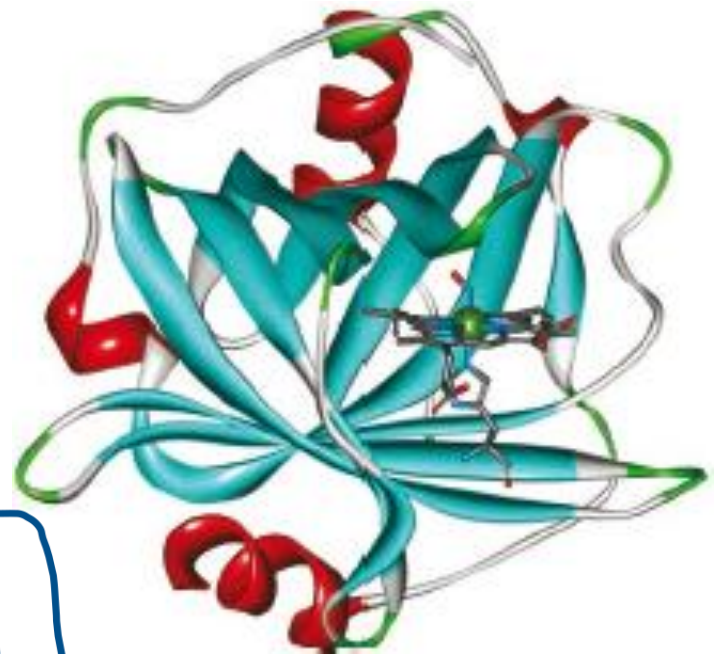




NO bound to NP1 (NP1-NO)



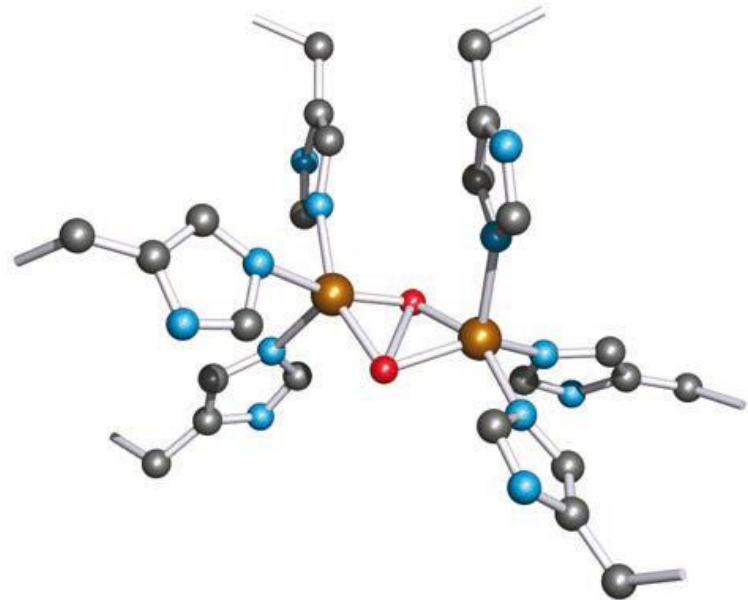
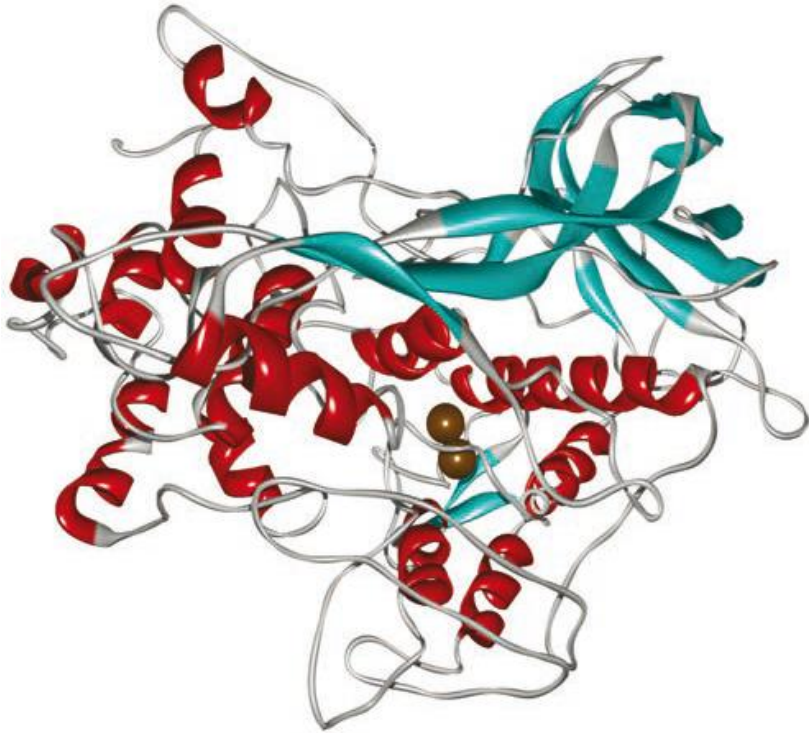
Histamine

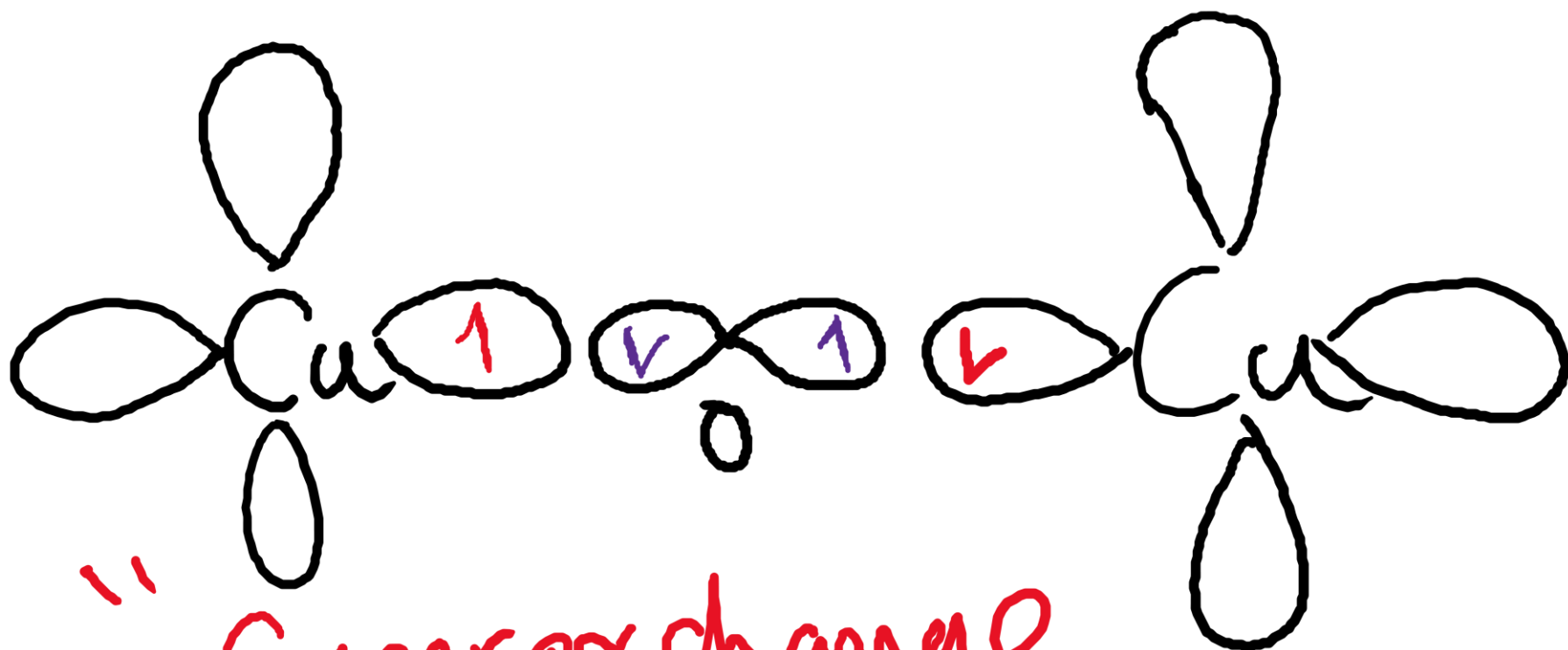


NOT FOR EXAM

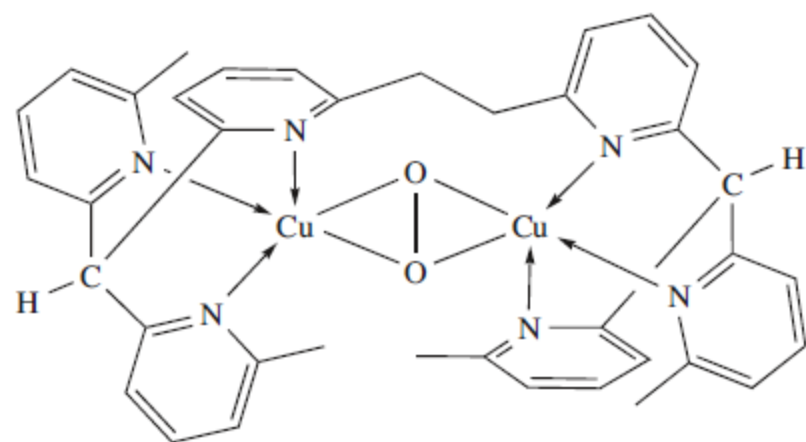
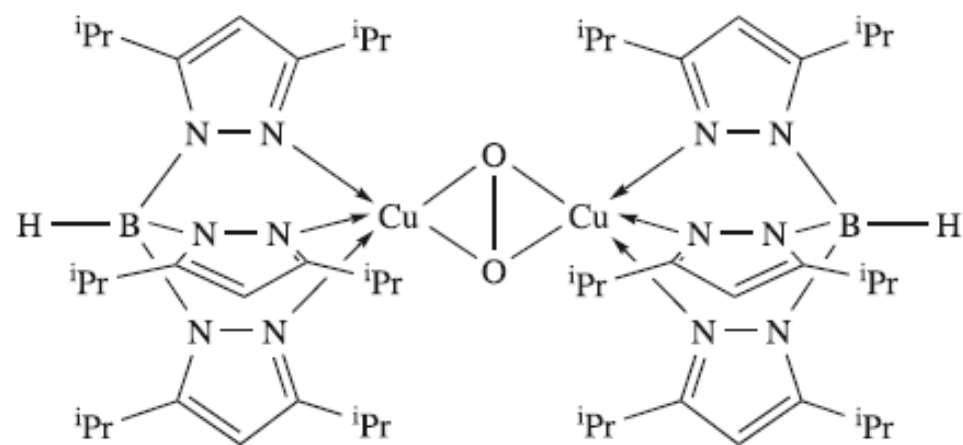
Haemocyanins

O₂-carrying copper-containing proteins in molluscs (e.g. whelks, snails, squid) and arthropods (e.g. lobsters, crabs, shrimps, horseshoe crabs, scorpions)





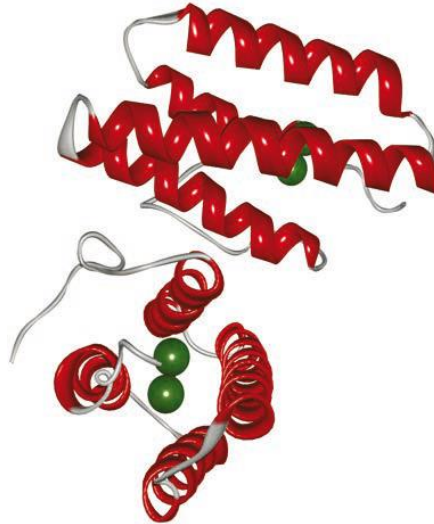
" Superexchange



Haemerythrin

In marine invertebrates such as annelids (segmented earthworms), O_2 is transported by haemerythrin, a non-haem Fe-containing protein

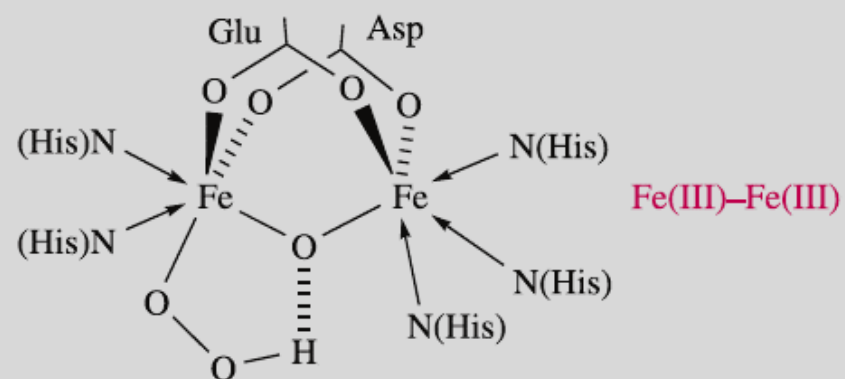
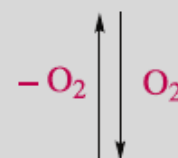
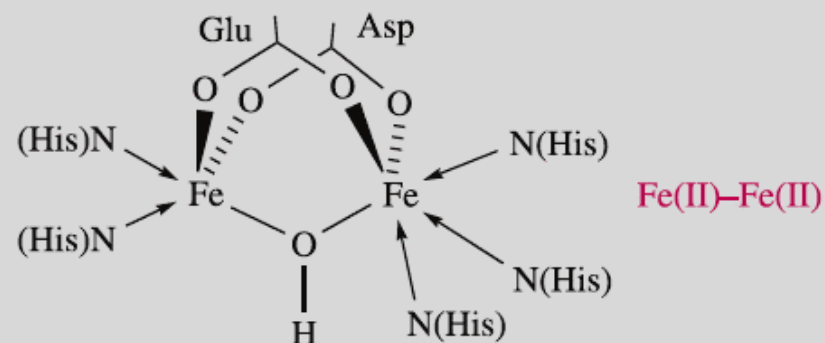
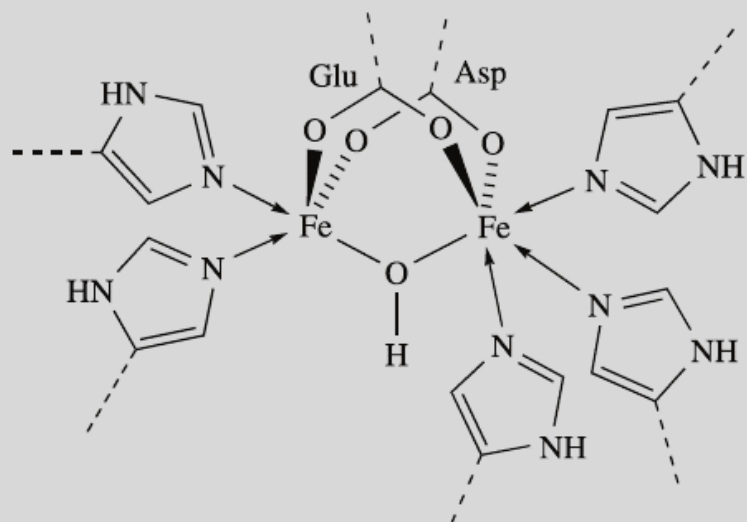
(Mr wt ~ 108 000)

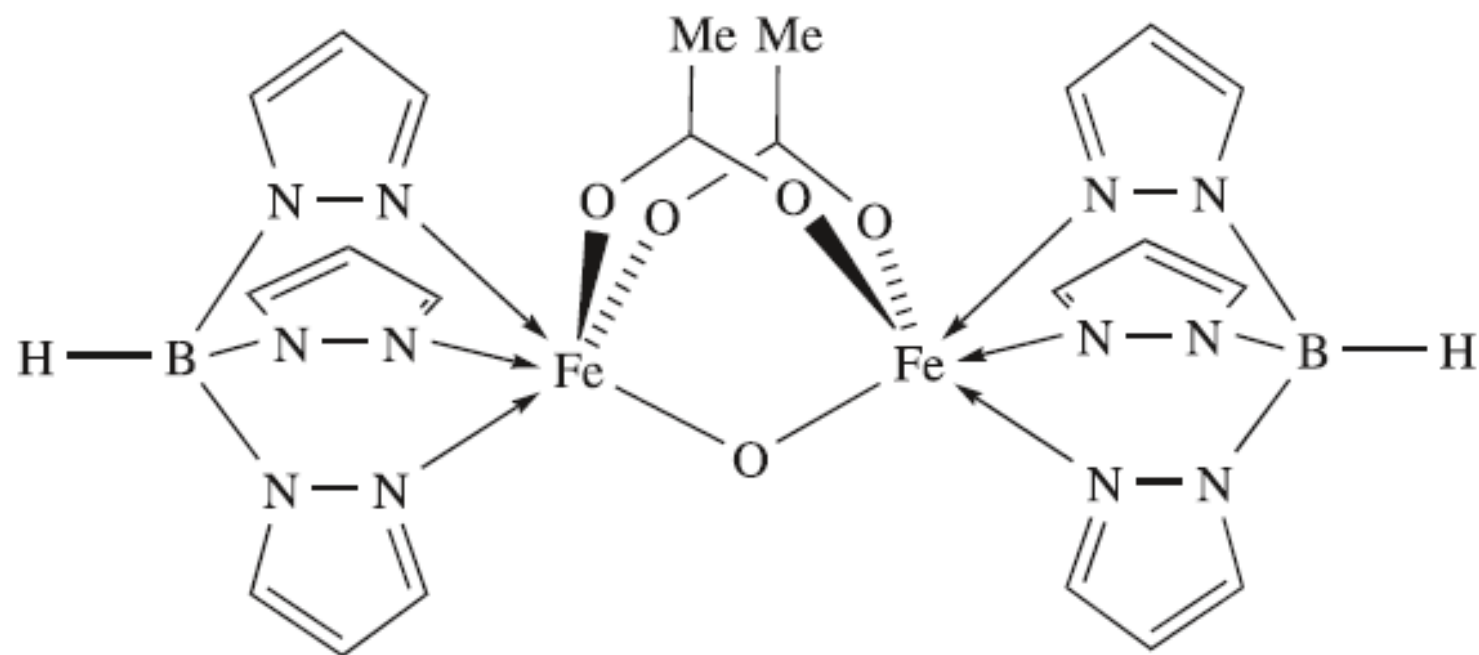


In the deoxy-form, a hydroxido-bridged $[Fe(II)]_2$ unit is present.

The two $Fe(II)$ centres in deoxyhaemerythrin are strongly antiferromagnetically coupled through the Fe-O-Fe bridge

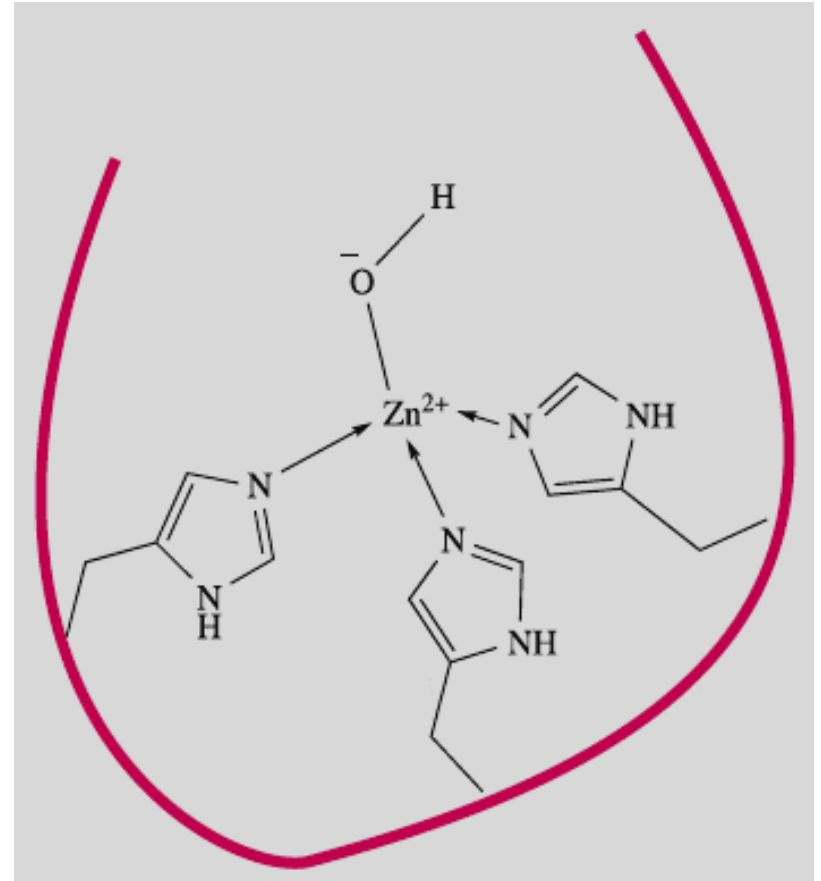
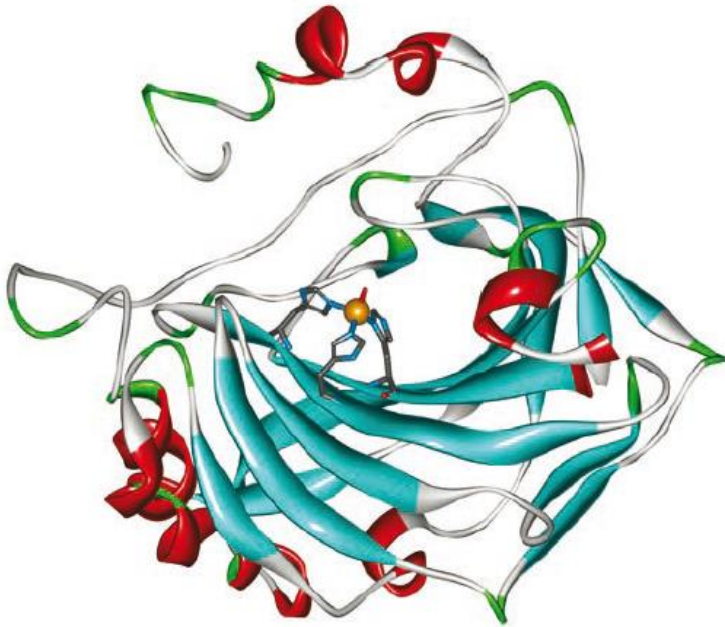
Unlike haemoglobin, haemerythrin exhibits no **cooperativity** between the sub-units during O_2 binding

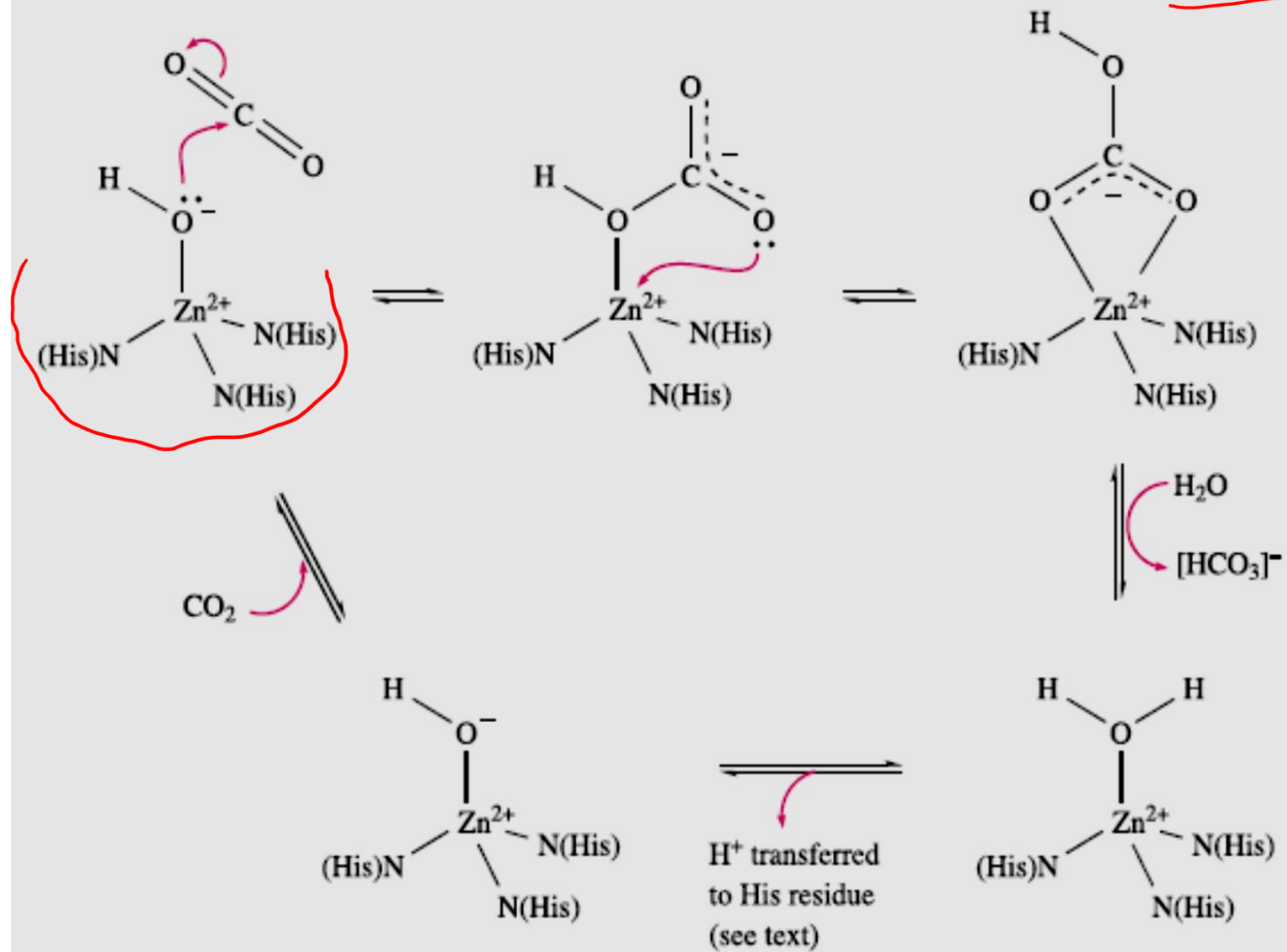


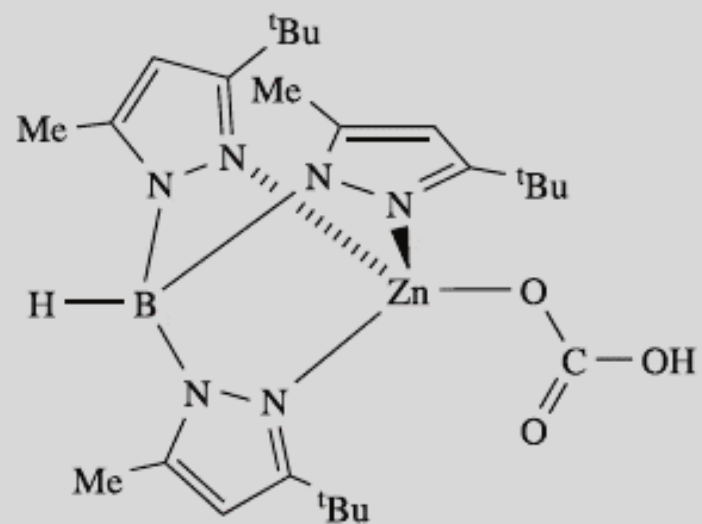
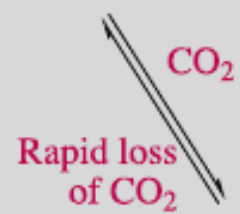
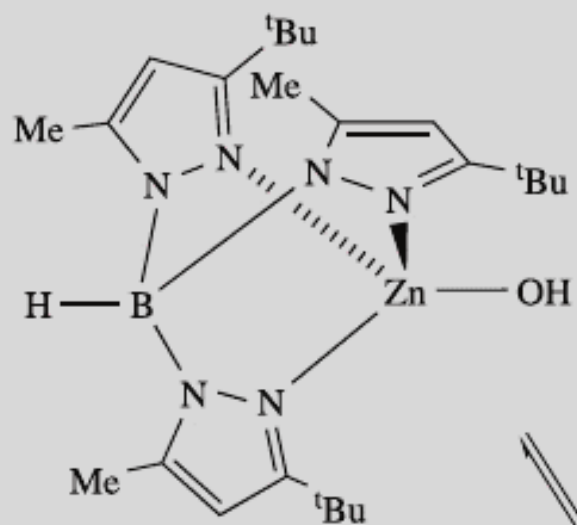


CARBONIC ANHYDRASE

The metalloprotein consists of 260 amino acids







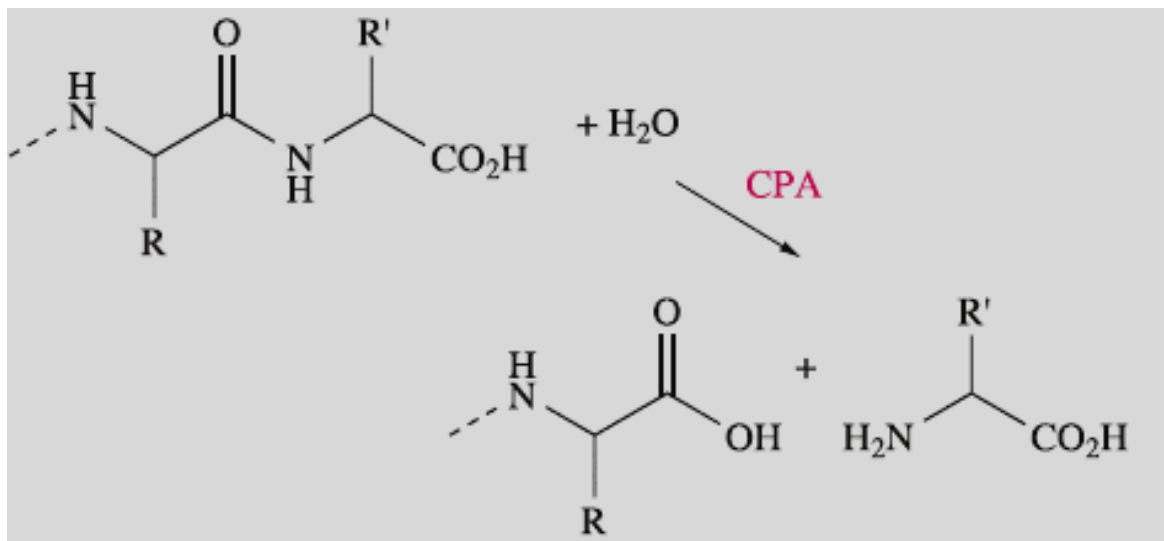
Carboxypeptidase

Metalloenzyme which catalyses the cleavage of a peptide link in a polypeptide chain

Carboxypeptidase A is monomeric (Mr 34 500)

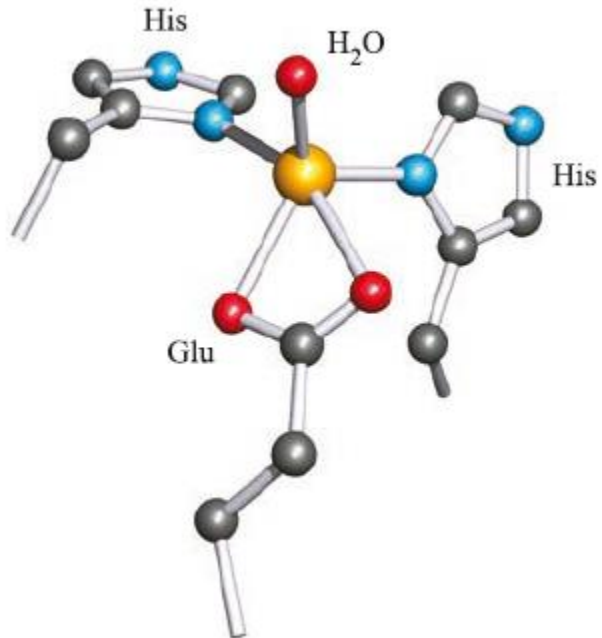
Near the surface of the protein lies a pocket in which a Zn^{2+} ion is bound to the protein backbone by one bidentate Glu and two His residues.

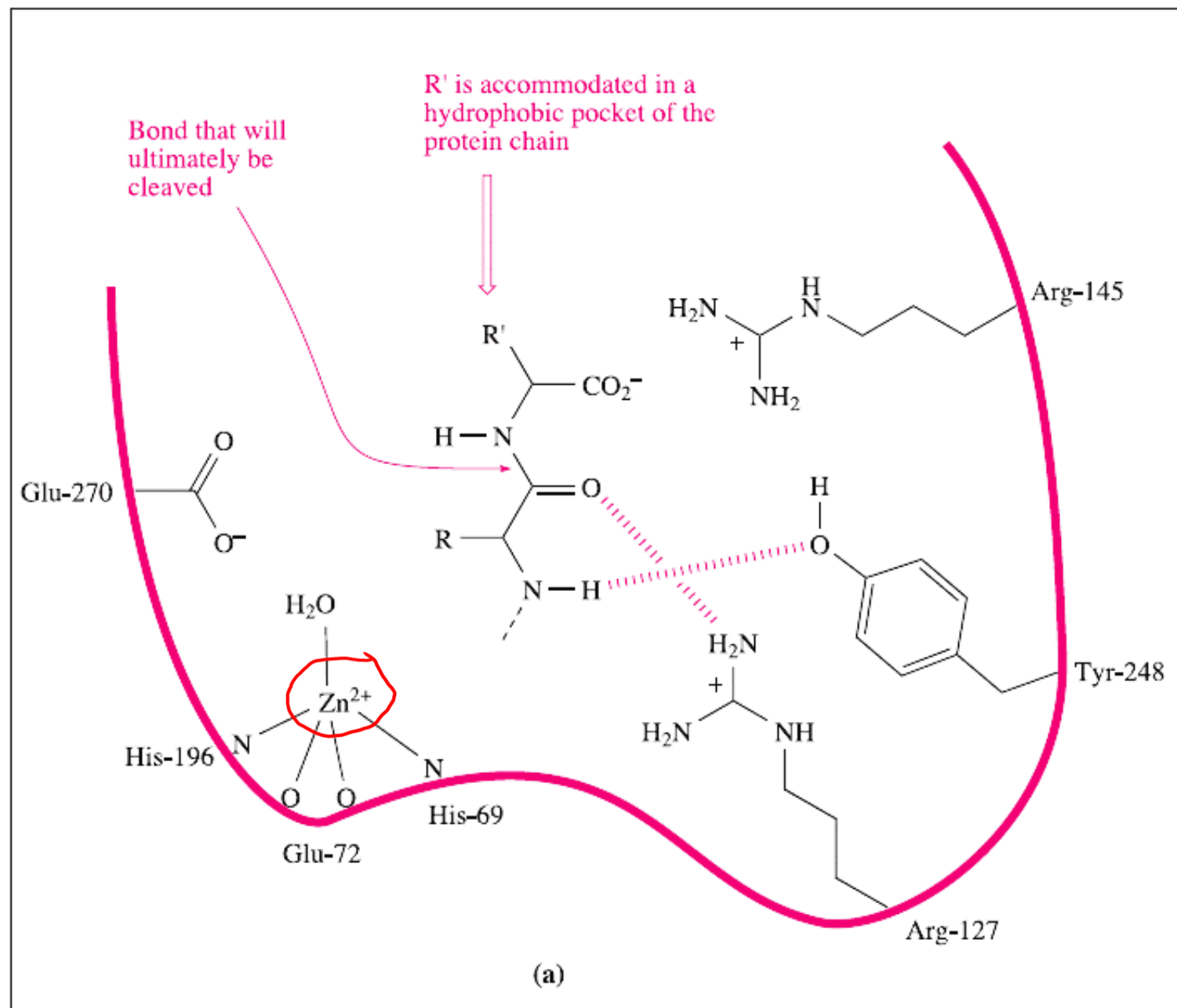
A 5-coordinate coordination sphere is completed by a water molecule



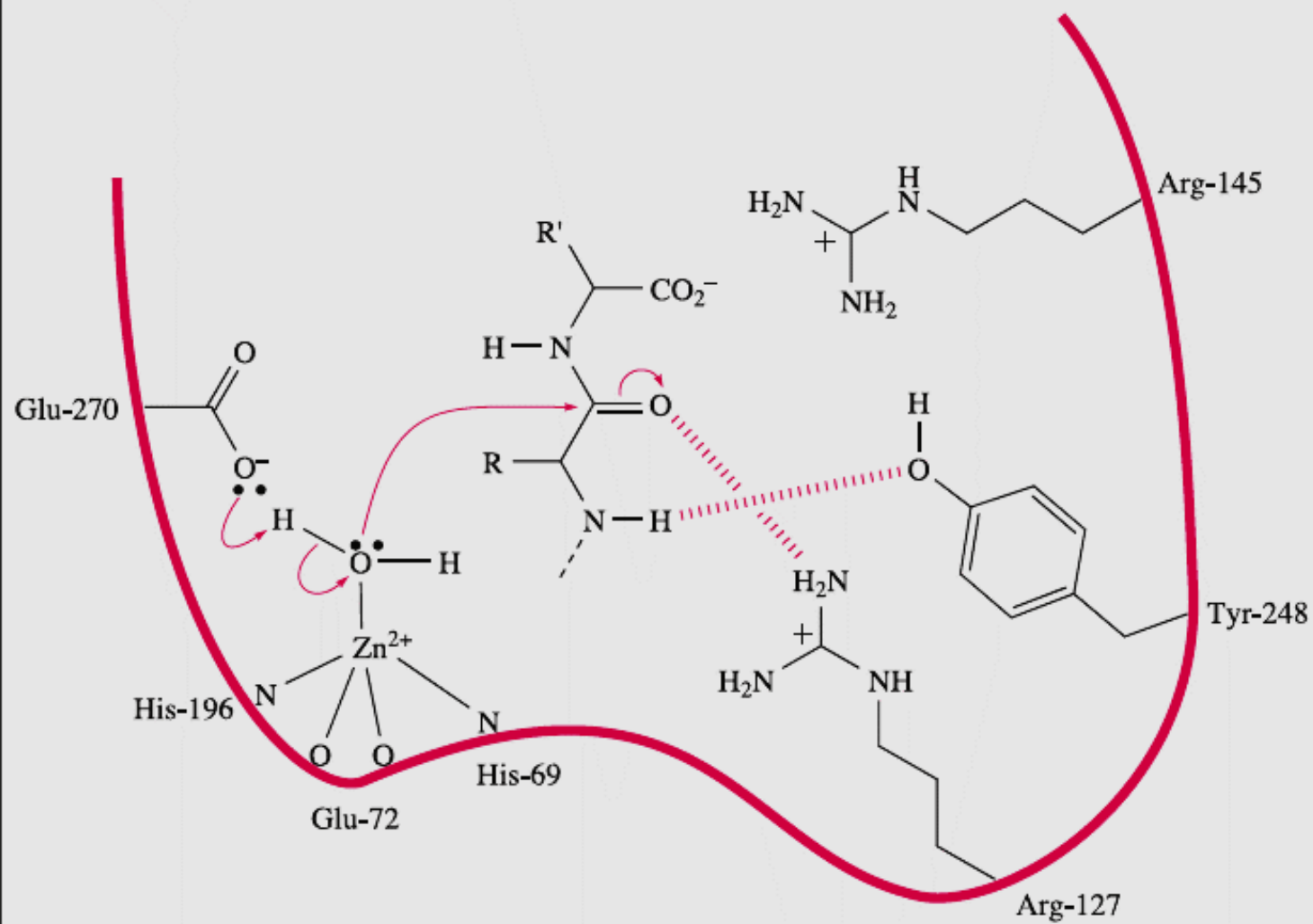
Carboxypeptidase

Active Site



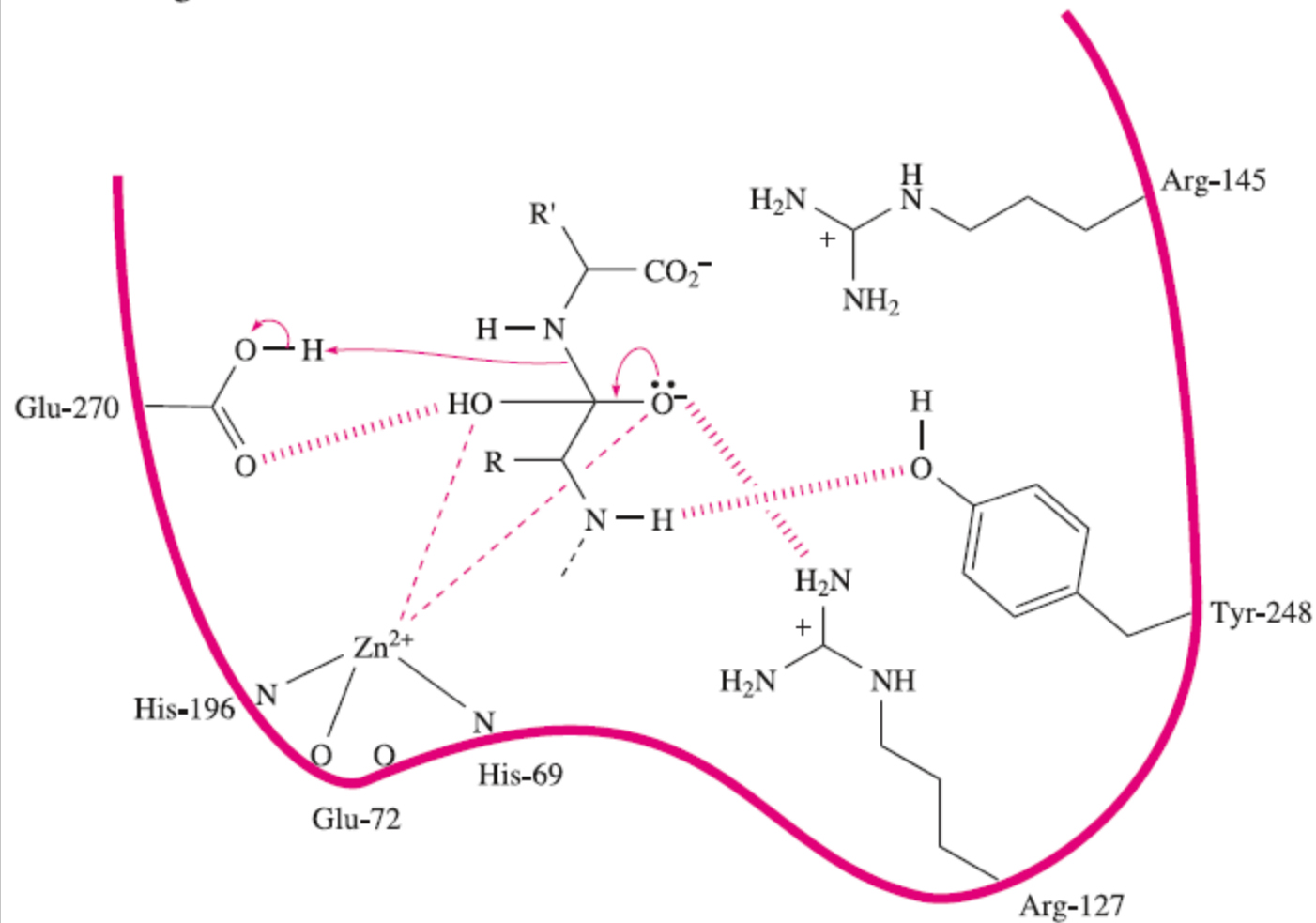


Nucleophilic attack on carbonyl C atom



(b)

Cleavage of the C–N bond



(c)